

TJ AIR Research Project Proposal

- **Title**

- Brain Tumor MRI Segmentation with U-NET Architecture

- **Members**

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- **Introduction**

- **Paragraph 1: Broad Problem + Why It Matters**

- Brain tumors are categorized as the abnormal growth of cells in the brain, which can be either benign (non-cancerous) or malignant (cancerous). Approximately 10-20 thousand individuals in the United States pass away annually from the fatal effects of Glioblastoma, the most common type of brain tumor. U-NET architecture can prove to be a valuable medical tool since it is a more specialized, precise convolutional neural network that is extremely practical in the medical field. Recent studies have shown detection rates of 98-99% with this framework, showing great potential in saving lives. By creating an AI computer vision model that utilizes this unique class of image recognition to predict brain tumors before they become symptomatic, doctors can prescribe early treatment to victims, increasing the effectiveness of recovery and saving the lives of many.

- **Paragraph 2: Short Literature Review**

- Several studies have explored deep learning approaches for brain tumor segmentation in MRI images. Ronneberger et al.

(2015) specifically introduced U-NET architecture for biomedical image segmentation, using a contracting path to capture context and a symmetric expanding path for precise localization. Walsh et al. (2022) proposed a lightweight U-NET architecture trained on the BITE dataset using three perspective planes instead of 3D volumes achieving a mean IoU (intersection-over-union) of 89% without requiring data augmentation or large amounts of training data. Preetha et al. (2025) combined a Multi-Scale Attention U-NET with EfficientNetB4 encoder on the Figshare brain tumor dataset, achieving 99.79% accuracy, a Dice coefficient of 0.9339 and an IoU of 0.8795 by using multi-scale attention mechanisms (3x3, 5x5, and 7x7 kernels) to capture tumor boundaries at different resolutions while maintaining low computational overhead. Nguyen et al. (2024) proposed a 3D U-NET model combined with a context transformer (CoT) to capture long-range spatial dependencies in MRI scans, training on the BraTS2019 dataset and achieving Dice scores 82.0%, 81.5%, and 89.0% for enhancing tumor, tumor core, and whole tumor segmentation.

- **Paragraph 3: Your Project's Contribution**

- Previous research on brain tumor MRI analysis(above) has shown that conventional neural networks, and in specific U-net architecture, work well for medical image segmentation because they capture all the details and provide good results. However, there are many limitations to these studies. Many existing methods rely on large and perfect data sets that require perfection, which is unrealistic to acquire. Many also have trouble detecting unusually shaped tumors and have trouble generalizing across different MRI modalities. We want to address these limitations by implementing and evaluating a U-Net based model which is designed to improve segmentation and tumor boundary detection consistency along MRI slices. We will implement clinically-relevant data and will build and evaluate U-Net architecture to do so.

● **Benchmarks**

- **Benchmark by End of February (Intermediate Milestone)**
 - By this benchmark we plan to filter all of our data and run a simple baseline model with our designed U-net architecture. If the model is successful with basic tests, we will also try to implement a 3D U-Net which will allow it to process more advanced data sets such as volumetric MRI scans
- **Benchmark by Mid-April (Spring Milestone)**
 - At this point we hope to be completely finished with our model and we should start transitioning into a more testing heavy phase with various datasets. We want our model to perform with at least 85% accuracy in detecting brain tumors. In addition, we also want to try to implement *Attention U-Net*. This will help in precision suppressing irrelevant background features.
- **Benchmark by End of May (Main Goal)**
 - The final objective of this project is to create a working U-Net model that can effectively recognize tumor features. We also want to make sure that the model can capture even the most distinct parts of the tumor without fail and output a segmentation map where each pixel gets its own labeling as either tumor or healthy.
- **Dream Goal (Aspirational Stretch Goal)**
 - We aim to optimize the 3D U-Net architecture for real-time inference on edge-computing devices. This would involve significantly reducing the model's parameter count (via pruning or quantization) so that it could theoretically provide instant segmentation feedback to a radiologist during an active scan, rather than requiring post-processing on a server.

● **Research Procedure & Timeline (Hard Deadlines)**

- Week 1:
 - Finalize literature review on U-net and 3D U-Net Architecture
- Week 2:
 - Download our MRI Datasets from Kaggle

- Filter for any outliers we deem unnecessary
- Week 3:
 - Implement our 2D U-Net Architecture and train our model with previous MRI slices to establish our baseline models performance metrics
- Week 4:
 - Improve our baseline U-Net model
 - Tune hyperparameters such as learning rate and batch size
 - Improve our tumor boundary consistency
- Week 5:
 - Transition and implement into our 3D U-Net model for multi-dimensional MRI segmentation
 - Compare performance between the 2D and 3D U-Net models
 - Fine tune consistency and performance based on results if needed
- Week 6:
 - Evaluate our model based on our IoU, Dice Coefficient, and our Accuracy Percentage
 - Test using generalized MRI datasets
 - Analyze results for any false positives, and box loss
- Week 7:
 - Implement Attention U-Net in order to suppress irrelevant background features
 - Compare model against the baseline and 3D U-Net Models
- Week 8:
 - Finalize model architecture and training
 - Create segmentation maps and performance graphs (loss graphs, Dice and IoU Comparisons)
- Week 9:
 - Begin sections of the research paper (Methodology, Results, and Discussion)
 - Use figures from week 8 to reference back to and make graphs relating to our comparisons between models
 - Draft presentation slides explaining impact, our novel contribution, and model design

- Week 10:
 - Finish final report
 - Prepare for presentation speaking portions and Q&A along with demonstrations

● References

Ronneberger, Olaf, et al. "U-Net: Convolutional Networks for Biomedical Image Segmentation." *SpringerLink*, Springer International Publishing, 1 Jan. 1970, link.springer.com/chapter/10.1007/978-3-319-24574-4_28.

R, Preetha, et al. "Brain Tumor Segmentation Using Multi-Scale Attention U-Net with EFFICIENTNETB4 Encoder for Enhanced MRI Analysis." *Nature News*, Nature Publishing Group, 22 Mar. 2025, www.nature.com/articles/s41598-025-94267-9.

T.Nguyen¹, Thien-Qua, and ¹University of Information Technology. "Brain Tumor Segmentation in MRI Images with 3D U-Net and Contextual Transformer." *Brain Tumor Segmentation in MRI Images with 3D U-Net and Contextual Transformer*, arxiv.org/html/2407.08470v1. Accessed 23 Jan. 2026.

Walsh, Jason, et al. "Using U-Net Network for Efficient Brain Tumor Segmentation in MRI Images." *ScienceDirect*, Nov. 2022, www.sciencedirect.com/science/article/pii/S2772442522000429.